Dai Yuxuan
3036292116
CompSc (minor in math)

**MATH 2241 INTRODUCTION TO MATHEMATICAL ANALYSIS, FALL 2026**
**ASSIGNMENT 2**

Due date: September 28 (Monday), 22:00 pm.

(1) Let $(a_n)$ be a sequence of positive integers. Define

$$x_1 = a_1, \quad x_2 = a_1 + \frac{1}{a_2}, \quad x_3 = a_1 + \frac{1}{a_2 + \frac{1}{a_3}}, \quad x_4 = a_1 + \frac{1}{a_2 + \frac{1}{a_3 + \frac{1}{a_4}}},$$

and so on.

(a) Show that the subsequences $(x_{2n})$ and $(x_{2n-1})$ converge.

(b) Suppose that $a_n = 1$ for all $n$. Determine whether $(x_n)$ converges. Find $\lim_{n\to\infty} x_n$, if it exists. (Hint: You may use [BS, Section 3.4, Exercise 6], or see the discussion from the lecture on September 17.)

(2) Let $A \subset \mathbb{R}$. An element $x \in \mathbb{R}$ is said to be a cluster point of $A$ if there exists a sequence $(x_n)$ in $A$ such that $(x_n)$ converges to $x$. Note that $x$ is not necessarily an element of $A$.

(a) Show that an element $x \in \mathbb{R}$ is a cluster point of $A$ if and only if there exists a sequence $(x_n)$ in $A$ such that

$$|x_n - x| < \frac{1}{n}$$

for all $n$.

(b) Let $A \subset \mathbb{R}$ be nonempty and bounded. Let $\mathcal{M}$ be the set of all cluster points of $A$. Show that the maximum of $\mathcal{M}$ exists.

Reference Book: [BS] Robert G. Bartle and Donald R. Sherbert, *Introduction to Real Analysis*, Wiley, 2011, Fourth Edition.

P1

(a)

① We want to show that $(x_{2n})$, $(x_{2n-1})$ are monotone

Let $f(i,j) = a_i + \cfrac{1}{a_{i+1} + \cfrac{1}{\cdots \cfrac{1}{a_j}}}$

$\therefore x_n = f(1,n) = a_1 + \cfrac{1}{f(2,n)} \quad (n>1)$

$\therefore x_{2n+2} < x_{2n}$

$\Leftrightarrow f(1,2n+2) < f(1,2n)$

$\Leftrightarrow a_1 + \cfrac{1}{f(2,2n+2)} < a_1 + \cfrac{1}{f(2,2n)}$

$\Leftrightarrow (-1) f(2,2n+2) < (-1) f(2,2n)$

$\Leftrightarrow (-1)^{2n-1} f(2n,2n+2) < (-1)^{2n-1} f(2,2n)$

$\Leftrightarrow a_{2n} + \cfrac{1}{a_{2n+1} + \cfrac{1}{a_{2n+2}}} > a_{2n}$ ✓.

$\therefore x_{2n+2} > x_{2n}$.

By same reasoning, we can prove that $x_{2n+3} < x_{2n+1}$.

$\therefore (x_{2n})$ are increasing, $(x_{2n+1})$ are decreasing.

② Note that $f(2,n) \geq 1 \quad \forall n > 2$

$\therefore \forall n > 2$, $x_n = a_1 + \cfrac{1}{f(2,n)} \leq a_1 + \frac{1}{1} = a_1 + 1$

and $a_1 < a_1 + 1$

$\therefore \forall n$, $a_1 < x_n \leq a_1 + 1$, i.e. $x_n$ is bounded.

$\therefore (x_{2n})$, $(x_{2n+1})$ are all bounded.

By the MCT, $(x_{2n}), (x_{2n+1})$ are all convergent.

(b) $\because a_n = 1 \quad \forall n$

$\therefore f(i,j) = f(i',j') = x_{j-i}$ iff $j-i = j'-i' > 0$

$x_n = 1 + \frac{1}{x_{n-1}}$

① We first show that $(x_n)$ is convergent.

$$\frac{|x_{n+1} - x_n|}{|x_{n+2} - x_{n+1}|} = \left| \frac{1 + \frac{1}{x_n} - x_n}{1 + \frac{1}{1+\frac{1}{x_n}} - (1 + \frac{1}{x_n})} \right|$$

$$= \left| \frac{\frac{x_n^2 - x_n - 1}{x_n}}{\frac{x_n^2 - x_n - 1}{x_n(x_n+1)}} \right|$$

$$= |x_n + 1| \in (2, 3) \quad (\text{from P1 (a)②})$$

i.e. $|x_{n+2} - x_{n+1}| \leq \frac{1}{3} |x_{n+1} - x_n|$

$(x_n)$ is contractive.

$\forall \varepsilon > 0$, $\forall N(\varepsilon) < n < m$.

$$|x_m - x_n| < \sum_{i=0}^{m-n-1} |x_{n+i} - x_{n+i-1}|$$

$$\leq \sum_{i=0}^{m-n-1} \cdot (\tfrac{1}{3})^i \cdot (\tfrac{1}{3})^{n - N(\varepsilon)} \cdot (\tfrac{1}{3})^{N(\varepsilon)} \cdot |x_2 - x_1|$$

$$\leq (\tfrac{1}{3})^{N(\varepsilon)} \cdot |x_2 - x_1| = (\tfrac{1}{3})^{N(\varepsilon)}$$

So we set $N(\varepsilon) = \left\lceil -\frac{\ln 3}{\ln \varepsilon} \right\rceil$, then $\forall m > n > N(\varepsilon)$, $|x_m - x_n| < \varepsilon$.

$\therefore (x_n)$ is cauchy i.e. $(x_n)$ is convergent.

Note that $x_{n+1} = 1 + \frac{1}{x_n}$.

Suppose $\lim_{n\to\infty} x_n = L$.

$\therefore L = 1 + \frac{1}{L}$ i.e. $L = \frac{1 \pm \sqrt{5}}{2}$.

$\because x_n \in (1, 2)$

$\therefore L \in (1, 2)$

i.e. $\lim_{n\to\infty} x_n = \frac{1+\sqrt{5}}{2}$

P2.

(a)

① ($\Rightarrow$) Suppose $\exists (x_n) \subset A$, $\lim_{n\to\infty} x_n = x$

$\therefore \forall \varepsilon > 0$, $\exists N(\varepsilon) \in \mathbb{N}^+$, $\forall n > N(\varepsilon)$, $|x_n - x| < \varepsilon$.

So $\forall k > 0$, $k \in \mathbb{N}^+$, let $n_k = N(\frac{1}{k}) + 1$. then

$|x_{n_k} - x| < \frac{1}{k}$

i.e. $\exists (x_{n_k}) \subset A$ satisfy $|x_{n_k} - x| < \frac{1}{k}$

② ($\Leftarrow$) Suppose $\exists (x_n) \subset A$, $\forall n$ $|x_n - x| < \frac{1}{n}$.

then $\forall \varepsilon > 0$, Let $N(\varepsilon) = \lceil \frac{1}{\varepsilon} \rceil$.

$\forall n > N(\varepsilon)$, $|x_n - x| < \frac{1}{n} < \frac{1}{N(\varepsilon)} < \varepsilon$.

i.e., $x$ is a cluster point of $A$.

(b) $\because A \subset \mathbb{R}$ and $A$ is bounded

$\therefore$ By the completeness of $\mathbb{R}$, $\exists \sup A$

We first prove that $\exists m \in M$, $m \leq \sup A$.

Suppose $\exists m' \in M$, $m' > \sup A$.

then $\forall a \in A$, $|a - m'| < |\sup A - m'|$

i.e. Let $\varepsilon = |\sup A - m'|$, $\nexists a \in A$, $|a - m'| < \varepsilon$.

Case 1: $\sup A \in A$.

Then just let $x_n = m$, $\forall n \in \mathbb{N}^+$.

i.e. $\sup A \in M$, which is the maximum of $M$.

Case 2: $\sup A \notin A$.

Suppose $\exists \varepsilon > 0$, $\forall a \in A$, $|a - \sup A| > \varepsilon$.

$\because \forall a \in A,\ a < \sup A$.

then $\forall a \in A,\ a < \sup A - \varepsilon$. Otherwise, $|a - \sup A| \leq \varepsilon$.

$\therefore \forall n \in \mathbb{N}^+,\ \exists a \in A,\ |a - \sup A| < \frac{1}{n}$

We denote such $a$ as $x_n$.

According to $P_2$ (a), which means $\sup A$ is a cluster point.

i.e. $\sup A \in M$. $\rightarrow \sup A = \max M$

Therefore, maximum of $M$ exists, is $\sup A$.